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SUMMARY

AI systems engineer across the full stack, silicon to agents. Optimizes CUDA/Triton kernels and profiles LLM inference at the bottom (**215 TFLOP/s**, ~69% of A100 fp16 peak). Designs autonomous multi-agent systems, retrieval, and evaluation on top.

SKILLS

- **GPU & Performance:** CUDA, Triton (kernel DSL), PyTorch, kernel fusion, mixed precision (fp16), NVIDIA Nsight Systems/Compute, roofline/memory-bandwidth analysis, quantization, gem5, McPAT
- **AI Systems:** LLM inference profiling, FlashAttention/SDPA, KV-cache decode, multi-agent orchestration, MCP server design, Claude Code / Agent SDK, agent evaluation
- **Retrieval & Memory:** Hybrid search, reciprocal-rank fusion, cross-encoder reranking, sqlite-vec, SQLite FTS5
- **Languages & Infra:** Python, C++, CUDA, Docker, CI/CD, Slurm

EDUCATION

University of Wisconsin–Madison
M.S. Electrical and Computer Engineering

Madison, USA
Sep 2024 – Dec 2025

- Relevant Coursework: High Performance Computing, Advanced Computer Architecture, Machine Learning, Fault-Tolerant Computing

Vellore Institute of Technology
B.Tech in Electronics and Communication Engineering

Amaravati, India
2020 – 2024

PROJECTS

Triton vs cuBLAS LLM Kernel Benchmarking
Triton, CUDA, PyTorch, NVIDIA Nsight

Dec 2025 – Present
GitHub

- Fused linear+bias+**GELU** into one **Triton** kernel, cutting p50 latency **up to 1.73×** versus the **unfused** path (cuBLAS GEMM + separate bias/GELU launches) on small-batch memory-bound FFN projections. The win is removing an **HBM** round-trip and two launches, not a faster GEMM (median ~0.96× at larger batch).
- Benchmarked **76 LLM-shaped** GEMM kernels across Triton and cuBLAS on an **A100** (p50/p90 latency, throughput, bandwidth, tail jitter) to map where fusion wins. The hand-tiled Triton GEMM sustains **215 TFLOP/s**, **69% of fp16 tensor-core peak** (cuBLAS reaches ~90% on the same shapes).

Edge LLM GPU Profiling on Jetson Orin Nano
CUDA, NVIDIA Nsight Systems, PyTorch, FlashAttention

Feb 2026
GitHub

- Profiled **attention** and **KV-cache decode** on a Jetson Orin Nano (edge GPU for local LLM-agent deployment), sustaining ~**15.4K tokens/sec** single-stream FlashAttention decode and ~**7.5 TFLOP/s** peak fp16 GEMM.
- Found per-token **kernel-launch overhead** dominating the decode inner loop in Nsight Systems, the regime CUDA Graphs targets, and measured fused **SDPA** at ~**5.6×** over the naive attention path.

bert: Long-Context Memory for AI Coding Assistants (MCP)

May 2026 – Present

Python, MCP, sqlite-vec, sentence-transformers

GitHub

- Built **bert**, a local plugin (**MCP server**) that gives AI coding assistants (Claude Code, Cursor, Codex) a searchable memory of an entire codebase via **hybrid retrieval** (dense vector + keyword search, reranked), pulling only the relevant files into the model's context.
- Root-caused a silent bug in the retrieval fusion that had pinned a held-out QA eval near-random (**0.10**) and restored it to **0.85**. Across **3 BEIR datasets** (scifact/nfcorpus/fiqa) the bge-base stack beats the published BM25 baseline on all three. On scifact the cross-encoder-reranked hybrid scores **0.745 nDCG@10 (+0.08)**.
- Demonstrated retrieval is the only viable path past the context window: on a **3M-token** corpus ($3\times$ a **1M** window) with a fixed held-out QA set, full-context is infeasible and truncation scores **0.00** while bert holds **0.75** at a flat **~3.3K** tokens per query.

Autonomous ML Research Lab (Multi-Agent System)

Mar 2026 – Present

Python, C17, Apple Metal, Claude API

GitHub

- Built an autonomous system of **31 role-specialized agents** that runs an end-to-end **ML research loop** on a single 18 GB laptop: the agents plan experiments, write and run the code, and review each other's results, with a human only setting direction.
- Their core deliverable is a **from-scratch deep-learning training engine** in **C17** (hand-written autodiff, a Mixture-of-Experts transformer, 4-bit QAT, Accelerate/AMX matmul) with **no PyTorch or TensorFlow**, plus hand-written **Apple Metal** GPU kernels built and unit-tested as ready-to-enable acceleration.
- Caught an agent forge 5 approvals, then designed the integrity layer that stopped it: pre-registration, phase gates, and **cryptographic sign-off verification**, with **zero recurrence across 23 runs**.

HIVE: Autonomous Multi-Agent Software Org

Mar 2026 – Present

Rust, libp2p, Python, Claude Code

- Built and run **HIVE**, an autonomous software org of **45 role-specialized agents** (Director, Evaluator, Red Team) that has completed **~203 plan-build-evaluate cycles** on a single product. Authored the governance layer that coordinates them: signed operator values, a tiered reversibility-based decision protocol, and JSON hand-off contracts between agents.
- Directed the agents to build a **62K-line Rust** peer-to-peer reasoning engine gated by **1,300+ tests**: a forward-chaining **Rete** inference engine with truth-maintenance, a **16-module differential-privacy** layer (Rényi-DP, k-anonymity), and a **libp2p** stack (Kademlia DHT, Gossipsub).

ML-Guided CUDA Kernel Configuration

2025

Python, PyTorch, CUDA, Slurm

GitHub

- Trained a PyTorch MLP runtime surrogate (12 GEMM/launch features) that predicts CUDA kernel execution time at $R^2 = \mathbf{0.96}$ (MAE 0.018 ms), replacing exhaustive autotuning with argmin-over-grid config selection.
- Validated predicted block/grid configs to **within ~5% of the exhaustively-measured optimum** (median across workloads), with benchmarking and CUDA event-timing automated on Slurm.

MI300X Memory-System Characterization & gem5 Modeling

2025

HIP, gem5, McPAT, Python

- Characterized the **AMD MI300X** memory hierarchy with HIP microbenchmarks (vL1D ~130, L2 ~230, HBM ~630–760-cycle latencies, plus **4.6–4.8 TB/s** write bandwidth) and calibrated full-system **gem5** to **within 5% MAPE** of measured vL1D latency on the dominant access stride, down from 89%.
- Compared in-order and out-of-order CPU models in gem5 across SAXPY/DAXPY/IAXPY for CPI, latency, and power, with **McPAT** stats-to-power extraction.